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SoftKnow: Market-Basket Analysis for We-Commerce

Final Project – B

Group D10

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1. Understanding the Dataset

The first step in our analysis was to review the raw dataset provided by We-Commerce and define the meaning of each attribute. This defines the role of each attribute, distinguishing between technical identifiers and business-relevant information. This step corresponds to **Project Item 1**.

1.1 Attribute Dictionary

The dataset consists of "events", where each row corresponds to a single visit action on the website. The attributes are defined as follows:

Field	Description
tracking_record_id	Unique identifier for each event recorded in the database. Each row is unique and corresponds to a single visit on the website.
date_time	Timestamp when the event occurred (Format: YYYY-MM-DD HH:MM:SS). Essential for ordering events within a session.
user_gui	Unique identifier for registered users. Empty if the visitor is not a registered user.
campaign_id	Identifier for a campaign associated with a product. Empty if no campaign is linked to the specific visit.
product_gui	Unique identifier for the product visited. **Note:** This field may contain non-product values (e.g., `/customer/account/login/`, `home`, `open`) and requires filtering for accurate analysis.
company	Name of the company providing the product.
link	URL of the webpage visited by the user.
tracking_id	Likely the encoding or type of tracking event (e.g., `ASCII`, `UTF-8`).
meio	Channel or medium of the visit.
ip	Visitor's IP address. Due to dynamic IP allocation, it is less reliable for unique visitor identification than cookies.
browser	Identification of the visitor's browser. Can retrieve device type, operating system, and browser version.
session_id	Unique identifier for the visitor session. A visitor may generate multiple sessions; a session expires after inactivity or browser closure.
referer	URL of the previously visited page (referring page).
cookie_id	Unique identifier stored in the visitor's browser. Primary attribute for tracking unique visitors across multiple sessions.

Table 1.1: Description of tracking dataset fields

2. Data Profiling and Aggregated View

To understand the dataset and validate data quality, we performed an initial statistical analysis. This section addresses **Project Item 3**.

2.1 General Statistics

Using SQL aggregation queries, we extracted the base metrics of the dataset:

- **Total Events:** The total count of rows in the ‘track‘ table, representing individual user interactions with the website.
- **Total Visitors:** Determined by counting distinct `cookie_id` values, providing an estimate of unique users in the dataset.

The dataset contains a total of 415,863 events generated by 263,137 distinct visitors, indicating that the majority of users interacted with the platform only briefly.

2.2 Distribution Analysis

We generated views to analyze the relationship between Visitors, Sessions, and Events.

2.2.1 Visitor Loyalty (Sessions per Visitor)

To understand how frequently users return to the platform, we analyzed the number of sessions per visitor. Each visitor is uniquely identified by `cookie_id`, and the number of distinct sessions was computed per visitor.

- **Goal:** Distinguish between one-time visitors and loyal, recurring customers.
- **Observation:** The vast majority of visitors have only a single session, while a very small number of users exhibit high recurrence, with more than 20 sessions. As shown in Figure 2.1, this indicates that while most users interact briefly with the platform, a small fraction of highly engaged users generate a large share of total sessions and interactions.

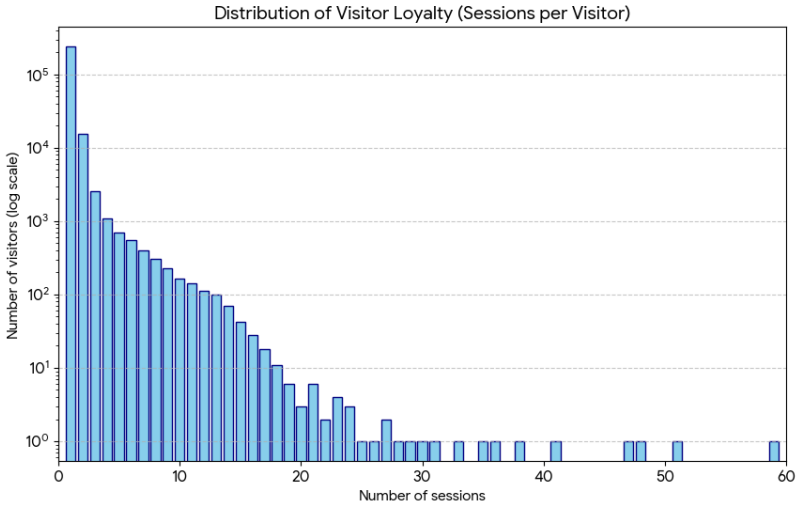


Figure 2.1: Distribution of Visitor Loyalty

2.2.2 Session Depth (Events per Session)

Then we examined the number of events per session. Each session is uniquely identified by `session_id`, and the number of events within each session was counted.

- **Goal:** Identify engagement levels within individual sessions.
- **Observation:** Very short sessions (1–2 events) suggest brief interactions, where users leave quickly after minimal engagement. Longer sessions may indicate highly engaged users, but extremely long sessions could also signal automated activity or atypical browsing patterns, as illustrated in Figure 2.2.

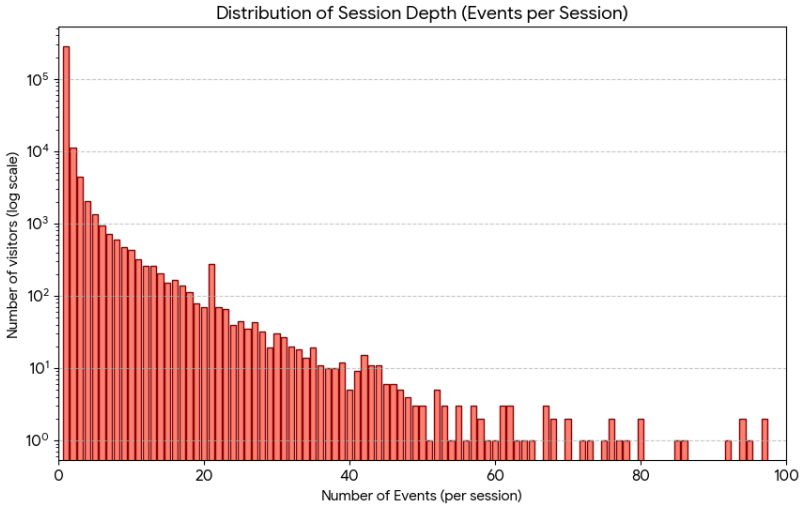


Figure 2.2: Distribution of Session Depth

[0]The graphics were generated using Gemini AI based on the data from the views. Chat for the AI-generated visualizations: `https://gemini.google.com/share/b1cc906230a8`.

3. Market Basket Analysis

This analysis aims to discover patterns of co-occurrence among products visited by users, a process known as **Market Basket Analysis (MBA)**, often used in retail and e-commerce to identify frequently bought or viewed products.

3.1 Problem Formulation

MBA treats each purchase or browsing session as a *transaction*, and the items in that transaction as *products*. Association rules derived from this reveal which products are often viewed together, which items can be grouped for promotions, and which combinations indicate user preferences or trends.

In our dataset (**Project Item 2**), these concepts are mapped as follows:

- **Transaction (Basket):** Each `session_id` corresponds to a single transaction. All products viewed within the same session are considered to be "bought together."
- **Item:** Each unique `product_gui` viewed during the session is treated as an item in the transaction.

Although transactions are defined at the session level, the `cookie_id` is essential for identifying the individual visitor.

3.2 Data Filtering

Given the large volume of events in the dataset, analyzing every single event is computationally expensive and may produce misleading results due to noise, such as bot activity or accidental clicks. To focus on meaningful user behavior, we applied a filtering step (**Project Item 4**) based on visitor activity:

> **We analyze only the subset of events generated by visitors (`cookie_id`) who have between 5 and 30 sessions (`session_id`).**

This filtering eliminates one-time visitors (fewer than 5 sessions) while also removing potential bots (more than 30 sessions) that could influence the rules. As illustrated in Figure 2.1, most visitors have fewer than 5 sessions. By applying these thresholds, each transaction corresponds to a session of a loyal, human user, improving the quality of the discovered patterns.

4. Data Preparation and Transformation

To prepare the dataset for Market Basket Analysis, we implemented a multi-stage data transformation process (**Project Items 5 and 6**). This ensures that each transaction is correctly defined, and items are consistently formatted.

4.1 Data Export from SQL

A custom SQL view (`v_export`) was created to join session-level statistics with the tracking data. Only visitors with 5–30 sessions were included. The exported data contains:

- `cookie_id` – identifies the visitor and is used for filtering and context.
- `session_id` – defines the transaction (one transaction per session).
- `product_gui` – defines the item in the basket.

Using PostgreSQL `COPY` command, the filtered data was exported as a semi-colon separated text file. Non-product events (e.g. 'home', '/customer/account/logoutsuccess/', ...) were removed in the SQL view.

4.2 Data Normalization

Even after filtering, the data may contain inconsistencies in product identifiers. In Python, each item was normalized by converting to lower case, removing accents and special characters, removing whitespaces and quotes, and replacing symbols that conflict with the basket format ('=' → '+').

4.3 Basket Generation

Normalized data was transformed into the '.basket' and '.tab' formats for Orange:

- Transactions are grouped by `session_id` (one basket per session).
- Each line in '.basket' lists all products viewed in that session.
- The '.tab' file represents the same data as a sparse matrix, with products as features and transactions as rows.

This process ensures efficient Market Basket Analysis with transactions reflecting user behavior and product co-occurrences while retaining visitor-level context via `cookie_id`.

5. Marketing Decisions from Rules

The objective of this exercise (**Project Item 7**) is to derive marketing insights from browsing data through a controlled exploration of association rules. To maintain interpretability and relevance, the analysis was limited to a maximum number of rules (**maxR**) and focused on selected ranges of support and confidence.

5.1 Association Rule Metrics

Association rules were evaluated using standard metrics: **Support** measures how frequently an itemset occurs, **Confidence** indicates the probability that the consequent appears given the antecedent, and **Lift** compares the observed co-occurrence of items to what would be expected if the items were independent. A lift greater than 1 indicates a positive association.

These metrics allow the identification of association rules that are both frequent and meaningful, balancing popularity with predictive strength.

5.2 Rule Generation Process

The dataset was represented in basket format and then one-hot encoded for association rule mining. Frequent itemsets were generated for predefined support ranges, and association rules were derived using different confidence thresholds. The number of rules was monitored: if a combination produced more than the predefined maximum (**maxR**), exploration for that combination was stopped. Only non-zero rule sets within the limit were retained, ensuring that the analysis focused on patterns relevant to marketing decisions.

5.3 Marketing Analysis and Decisions

5.3.1 Most Visited Products

High-support, low-confidence rules identified the platform's most visited products: **lon_4508**, **lon_2125**, **botins**, **pumpseopentoes**, **botas**, and **lon_4504**. These items drive the majority of user interactions and should be particularly featured on landing pages, category highlights, and promotional banners.

5.3.2 Products That Should Be Kept Together on a Web Page

Medium-to-high support combined with high confidence highlighted strong co-occurrence patterns. Key findings include:

- {**lon_4504**, **lon_2125**} → **lon_4508**
- {**lon_4004**, **lon_4504**} → **lon_4508**
- {**botas**, **pumpseopentoes**} → **botins** (confidence>0.91)

These patterns indicate two groups of products that are frequently visited together: the first includes the lon_ series products (**lon_4504**, **lon_2125**, **lon_4004**, **lon_4508**), and the second includes footwear products (**botas**, **pumpseopentoes**, **botins**). Marketing can use these insights for grouped recommendations, bundled displays, or "frequently viewed together" sections.

6. Visual Analysis and Reporting with Orange

This chapter addresses **Project Item 8** and focuses on visually presenting the association rules obtained in the market-basket analysis. The goal is to communicate the results clearly to We-Commerce rather than discovering new patterns.

6.1 Orange Workflow

The workflow processes the basket-formatted dataset generated earlier and includes the following widgets:

- **File**: Loads the transactional dataset in basket format.
- **Data Table**: Allows inspection of the imported transactions.
- **Frequent Itemsets**: Generates frequent itemsets using a predefined minimum support.
- **Association Rules**: Derives association rules based on minimum confidence and support thresholds.

This workflow enables interactive exploration of rules and metrics without additional coding.

6.2 Graphical Exploration of Rules

The Association Rules widget displays rules in a table with metrics such as support, confidence, lift, leverage, and coverage. Sorting and filtering by these metrics makes it easy to identify strong associations. The rules with higher confidence and lift values visually confirm the product groupings identified in Chapter 5.

6.3 Report Generation

Orange can export the workflow results as an HTML report, consolidating frequent itemsets and association rules. The report provides a clear summary of the strongest rules, interactive tables for inspection, and a visually accessible overview of the analysis. This approach ensures the results are informative and ready for business use by We-Commerce.